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Roll No

CE-7002-CBGS

B.E. VII Semester

Examination, June 2020

Choice Based Grading System (CBGS)

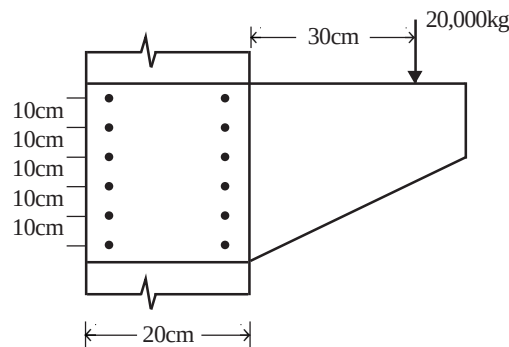
Structural Design -II (Steel)

Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.
iii) Use of relevant IS code is permitted.

1. A load of 20,000 kg is carried by a plate bracket riveted to a column as shown in Fig.1.1. Calculate the maximum force taken up by any rivot. 14



2. Design member of a truss is subjected to factored tension of 850kN. It consists of a channel ISMC 350@ 413N/m has to be connected to a 10mm thick gusset plate. The overlap length of the member with the gusset plate is limited to 300mm. Design the fillet welded connection assuming site welding. 14

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3. Write short notes on following: 14
 - i) Gantry Girder and its uses
 - ii) Plate Girder and its uses
 - iii) Plastic design of frames
4. Design a laced column 10m long to carry a factored axial load of 1000kN. The column is restrained in position but not in direction at both ends. Take two channels back to back with single lacing system and bolted connection. 14
5. Explain in detail the following : 14
 - i) Bracing for transmission towers
 - ii) Stability analysis of foundation for transmission towers
6. a) Draw neat sketches of various type of roof trusses. 7
b) What are the assumptions made in Euler's analysis? 7
7. A tie member $75\text{mm} \times 8\text{mm}$ connected to a 10mm thick gusset plate is to transmit a load 90kN. Design the fillet weld and calculate the necessary over lap. Assume site welding. 14
8. Design a purlin for a roof truss having the following data: 14
Span of the truss = 6m
Spacing of the truss = 3m c/c
Inclination of the roof = 30°
Spacing of purlin = 2m c/c
Wind pressure = 1.5kN/m^2
Roof coverage = A.C sheeting weighing 200N/m^2
Provide a channel section purlin.

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